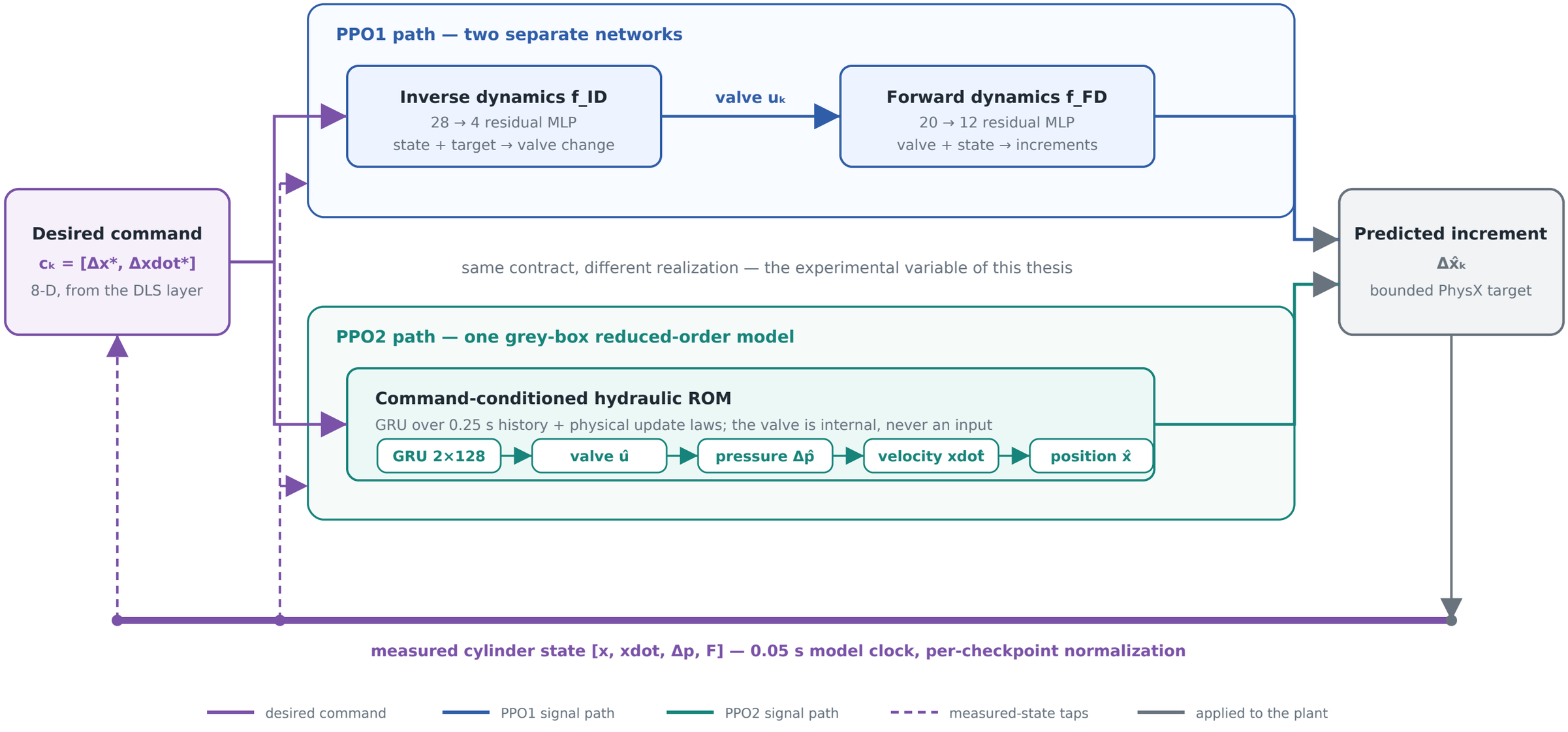
PPO1 path — two separate networks
Inverse dynamics f_ID
28 → 4 residual MLP
state + target → valve change
valve uₖ
Forward dynamics f_FD
20 → 12 residual MLP
valve + state → increments
Desired command
cₖ = [Δx*, Δxdot*]
8-D, from the DLS layer
same contract, different realization — the experimental variable of this thesis
Predicted increment
Δx̂ₖ
bounded PhysX target
PPO2 path — one grey-box reduced-order model
Command-conditioned hydraulic ROM
GRU over 0.25 s history + physical update laws; the valve is internal, never an input
GRU 2×128
valve û
pressure Δp̂
velocity xdot̂
position x̂
measured cylinder state [x, xdot, Δp, F] — 0.05 s model clock, per-checkpoint normalization
desired command
PPO1 signal path
PPO2 signal path
measured-state taps
applied to the plant